DUKAANSATHI AI

Voice-Powered Smart Kirana Assistant

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1. About My Project

I developed DukaanSathi AI to help small kirana shop owners manage their daily activities using voice commands. It supports Telugu and English, allowing shopkeepers to place customer orders without manually entering all the details.

2. Problem Statement

In many small kirana stores, shopkeepers manage customer orders, billing and inventory manually. This process takes time and may lead to mistakes. I wanted to develop a simple AI-powered application that makes these activities easier through voice interaction, especially for people who are comfortable speaking in their regional language.

3. Proposed Solution

DukaanSathi AI allows shopkeepers to place orders using Telugu and English voice commands. Gnani AI Speech-to-Text converts spoken commands into text. The application identifies customer names, product names, quantities and units, checks product availability and generates an order preview.

After the shopkeeper confirms the order, it is saved, billing is processed and inventory is automatically updated. Gnani AI Text-to-Speech provides voice-based order confirmation.

4. Key Features

- Telugu and English voice-based ordering.
- Customer name and product identification.
- Automatic quantity and unit extraction.
- Order preview and confirmation.
- Automated billing and inventory management.
- Low-stock and out-of-stock alerts.
- Alternative product suggestions.
- Duplicate-order protection.
- Sales analytics and revenue tracking.

5. Gnani AI Integration

- **Speech-to-Text (STT):** Used to convert Telugu and English voice commands into text.
- **Text-to-Speech (TTS):** Used to provide voice-based order confirmations.

6. Technologies Used

Python, FastAPI, SQLite, SQLAlchemy, HTML, CSS, JavaScript and Gnani AI APIs.

7. Target Users and Real-World Impact

This project is mainly designed for small kirana shop owners and local retailers. It helps simplify order processing, billing, inventory tracking and sales management. My aim is to make digital business operations easier and more accessible through regional-language voice interaction.

8. Project Link

GitHub Repository: https://github.com/Tejaswini8888/DukaanSathi-AI

Demo Video: https://drive.google.com/file/d/1sP9xELgGjvtzCK8u2OzuGM04nkEyU-ZT/view?usp=sharing